

Employee Payroll Management System (EPMS) - ERD

This document represents the Entity Relationship Diagram (ERD) for the Employee Payroll Management System (EPMS).

Database Relationships

Relationship	Description	Type
Department → Employee	One department has many employees	1:N
Department → Salary	One department manages many salary records	1:N
Employee → Salary	One employee receives many salary records	1:N

ADMIN

Column	Type
id	int (PK)
email	varchar(230)
password	int
created_at	int

DEPARTMENT

Column	Type
id	int (PK)
dep_code	varchar(12)
dep_name	varchar(254)
gross_salary	decimal(10,2)
total_deduction	decimal(10,2)
created_at	datetime
update_date	timestamp

EMPLOYEE

Column	Type
id	int (PK)
dep_id	int (FK)
dep_number	int
fname	varchar(255)
lname	varchar(255)

position	varchar(255)
address	varchar(255)
telephone	varchar(18)
gender	varchar(20)
hered_date	date

SALARY

Column	Type
id	int (PK)
dep_id	int (FK)
emp_id	int (FK)
net_salary	decimal(10,2)
month	date
create_at	timestamp

Recommended Improvements

- Use InnoDB engine instead of MyISAM to support foreign keys.
- Change password field type from int to varchar(255) for secure password hashing.
- Add foreign key constraints between related tables.
- Fix typographical errors such as hered_date → hired_date and cresated_at → created_at.

Normalized Relationship Flow

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Department
↓
Employee
↓
Salary

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